Problem Set 19

Areas Between Curves II

October 21, 2026

Problem 4

(a)

$$x^2 + 4x - 8x = 0 \implies x = \boxed{0 \text{ or } 4}$$

(b)

$$A = \int_0^4 2\sqrt{x}\,dx + \int_4^8 \sqrt{8x - x^2}\,dx = \boxed{\frac{32}{3} + 4\pi}$$

(c)

$$A = \int_0^4 \left(\sqrt{8x - x^2} - 2\sqrt{x}\right) dx$$

$$\int_0^4 \sqrt{8x - x^2}\,dx = \begin{bmatrix} u = x - 4 \\ du = dx \end{bmatrix} \int_{-4}^0 \sqrt{16 - u^2}\,du = 16 \int_{-\pi/2}^0 \cos^2\theta\,d\theta$$

$$= 8\theta + 8\sin\theta\cos\theta\Big|_{-\pi/2}^0 = 8\arcsin\frac{u}{4} + \frac{u\sqrt{16 - u^2}}{2}\Big|_{-4}^0$$

$$A = \boxed{4\pi - \frac{32}{3}}$$

Problem 5

(a)

$$a > b \implies \frac{1}{a^2} < \frac{1}{b^2}$$

$$\frac{x^2}{a^2} + \frac{y^2}{a^2} \le \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\boxed{x^2 + y^2 \le a^2}$$

$$x^2 + y^2 = a^2 \implies \boxed{y = 0, x = \pm a}$$

(b)

$$A_c = \pi a^2, A_e = \pi ab$$

$$\Delta A = \boxed{\pi a(a - b)}$$

(c)

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \implies y=\sqrt{b^2-\frac{b^2}{a^2}x^2}$$
$$y=\sqrt{a^2-x^2}-\sqrt{b^2-\frac{b^2}{a^2}x^2}=(1-\frac{b}{a})\sqrt{a^2-x^2}$$
$$\frac{x^2}{a^2}+\frac{y^2}{(a-b)^2}=1 \implies (1-\frac{b}{a})\sqrt{a^2-x^2}$$

Problem 6

$$y=\frac{x^2+3x+2}{x^2-3x+2}=0 \implies x=-1 \text{ or } -2$$
$$\int_{-1}^{-2}\frac{x^2+3x+2}{x^2-3x+2}\,dx=\int_{-1}^{-2}1+\frac{12}{x-2}-\frac{6}{x-1}\,dx$$
$$=x+12\ln|x-2|-6\ln|x-1|\Big|_{-1}^{-2}$$
$$=-1+24\ln 2-6\ln 3-12\ln 3+6\ln 2$$
$$=\boxed{30\ln 2-18\ln 3-1}$$

Problem 7

$$\frac{y^2}{4}=y+2 \implies y=2+2\sqrt{3}$$
$$\int_0^{4+2\sqrt{3}}2\sqrt{x}\,dx=\frac{4}{3}x^{3/2}\Big|_0^{4+2\sqrt{3}}=\frac{40}{3}+8\sqrt{3}$$
$$A=\frac{40}{3}+8\sqrt{3}-\frac{\pi}{2}-8-4\sqrt{3}$$
$$=\boxed{\frac{16}{3}+4\sqrt{3}-\frac{\pi}{2}}$$

Problem 8

(a)

$$A(c)=\int_0^c(f(c)-f(x))\,dx+\int_c^a(f(x)-f(c))\,dx$$

(b)

$$A(c)=c\cdot f(c)-\int_0^c f(x)\,dx+\int_c^a f(x)\,dx-f(c)\cdot(a-c)$$
$$A'(c)=c\cdot f'(c)-f(c)-f'(c)(a-c)+f(c)=\boxed{(2c-a)f'(c)}$$
$$f(x)=\sqrt{\arctan x} \implies f'(c)>0$$
$$2c-a=0 \implies \boxed{c=\frac{a}{2}}$$

(c) Given that the function is monotonically increasing, the value of $A'(x)$ only depends on the relative ratio of c to a, as demonstrated above.

Problem 9

(a) $g(x)$ exists as $f(x)$ is monotonically increasing.

$$f(0) = 0 + 0 + 1 = 1 \implies g(1) = \boxed{0}$$
$$f(3) = 27 + 9 + 1 = 37 \implies g(37) = \boxed{3}$$

(b)

$$\int_0^3 f(x)\,dx = \int_0^3 x^3 + 3x + 1\,dx$$
$$= \frac{x^4}{4} + \frac{3x^2}{2} + x\Big|_0^3 = \frac{81}{4} + \frac{27}{2} + 3 = \frac{147}{4}$$
$$A = 3 \cdot 37 - \frac{147}{4} = \boxed{\frac{297}{4}}$$

Problem 10

(a)

$$y = 4 - x^2 \implies x = -2 \text{ or } 2$$
$$\int_{-2}^2 4 - x^2\,dx = 4x - \frac{x^3}{3}\Big|_{-2}^2 = \frac{32}{3}$$
$$\int_{-\sqrt{4-b}}^{\sqrt{4-b}} 4 - x^2 - b\,dx = (4-b)x - \frac{x^3}{3}\Big|_{-\sqrt{4-b}}^{\sqrt{4-b}} = \frac{16}{3}$$
$$2(4-b)^{\frac{3}{2}} = \frac{16}{3} + \frac{2(4-b)^{\frac{3}{2}}}{3}$$
$$(4-b)^{\frac{3}{2}} = 4$$
$$b = \boxed{4 - 2\sqrt[3]{2}}$$

(b)

$$y = x - x^2 \implies x = 0 \text{ or } 1$$
$$\int_0^1 x - x^2\,dx = \frac{x^2}{2} - \frac{x^3}{3}\Big|_0^1 = \frac{1}{6}$$
$$kx = x - x^2 \implies x = 1 - k$$
$$\int_0^{1-k} (1-k)x - x^2\,dx = \frac{(1-k)x^2}{2} - \frac{x^3}{3}\Big|_0^{1-k} = \frac{(1-k)^3}{6}$$
$$\frac{(1-k)^3}{6} = \frac{1}{12}$$
$$k = \boxed{1 - \frac{1}{\sqrt[3]{2}}}$$

Problem 11

(a)

$$3x^3 - 2x + c = 0 \implies c = x_1, x_2$$

$$cx_1 - \int_0^{x_1} 2x - 3x^3\,dx = cx_1 - x^2 + \frac{3x^4}{4}\Big|_0^{x_1}$$

$$\int_{x_1}^{x_2} 2x - 3x^3 - c\,dx = x^2 - \frac{3x^4}{4} - cx\Big|_{x_1}^{x_2}$$

$$cx_1 - x^2 + \frac{3x^4}{4}\Big|_0^{x_1} = x^2 - \frac{3x^4}{4} - cx\Big|_{x_1}^{x_2}$$

$$c(x_1 - x_1 + x_2) = x^2 - \frac{3x^4}{4}\Big|_0^{x_2}$$

$$cx = x^2 - \frac{3}{4}x^4 \implies \frac{3}{4}x^3 - x + c = 0$$

$$3x^3 - 4x + 4c = 0 \implies 2x = 3c$$

$$\frac{81}{8}c^3 - 3c + c = 0$$

$$c = \boxed{\frac{4}{9}}$$

(b)

$$A_1 = \int_0^{x_1} c - f(x)\,dx$$

$$A_2 = \int_{x_1}^{x_2} f(x) - c\,dx$$

$$A_1 = A_2 \implies \boxed{\int_0^{x_2} (f(x) - c)\,dx = 0}$$

(c)

$$\int_0^{\pi-\alpha} \sin x - \sin\alpha\,dx = 0$$

$$-\cos x - x\sin\alpha\Big|_0^{\pi-\alpha} = 0$$

$$\cos\alpha - (\pi - \alpha)\sin\alpha + 1 = 0$$

$$\boxed{\cos\alpha + 1 = (\pi - \alpha)\sin\alpha}$$

$$\boxed{c \approx 0.725}$$